

Exploratory Data Analysis on the US Arrests Dataset Report

Introduction

This report presents an exploratory data analysis (EDA) of the US Arrests dataset, which contains crime statistics for all 50 US states from 1973. The purpose of this analysis is to understand patterns, distributions, and relationships between different types of crimes across the United States.

The dataset includes four key variables for each state:

- Murder: Number of murder arrests per 100,000 people
- Assault: Number of assault arrests per 100,000 people
- UrbanPop: Percentage of the state's population living in urban areas
- Rape: Number of rape arrests per 100,000 people

Description of the Dataset

The dataset contains exactly 50 rows, one for each state in the United States, no states are missing from the data.

Data types:

- City (object/string) – State names
- Murder (float64) – Murder arrest rates
- Assault (int64) – Assault arrest rates
- UrbanPop (int64) – Urban population percentage
- Rape (float64) – Rape arrest rates

First 5 rows of the dataset:

City	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
Carlifornia	9.0	276	91	40.6

Key statistics: Murder rates range from 0.8 to 17.4 per 100,000 (average 7.8), assault rates range from 45 to 337 per 100,000 (average 170.8), urban population ranges from 32% to 91% (average 65.5%), and rape rates range from 7.3 to 46.0 per 100,000 (average 21.2).

Data Cleaning

Before any analysis could begin, the dataset was checked for quality issues. The following steps were performed:

Step 1: Checking for missing values – Each column was examined for empty values. The results showed that all 50 states had complete data for every variable. No missing values were found.

Step 2: Checking for duplicate rows – The dataset was examined for any duplicate state entries. No duplicate rows were found, confirming that each state appears exactly once.

Step 3: Verifying data types – The data types were correct: murder and rape as decimal numbers, assault and urban population as whole numbers, and state names as text.

Step 4: Cleaning column names – Column names were stripped of any hidden spaces for consistency.

Since the dataset was already clean from the start, no additional cleaning steps were necessary.

Missing Data

The US Arrests dataset contained no missing values whatsoever. Every single one of the 50 states had complete information for all four variables: murder, assault, urban population, and rape.

Because there were no missing values, no special techniques such as deleting rows or filling in missing values were required.

Data Stories and Visualisations

This section presents the key findings from the analysis, supported by visualisations created using Python libraries Matplotlib and Seaborn.

Visualisation 1: Distribution of Crime Rates (Histograms)

Four histograms were created to show how each crime variable is distributed across the 50 states.

Murder Distribution: Murder rates range from 0.8 to 17.4 arrests per 100,000. Most states fall between 4 and 11, with a right-skewed distribution. Georgia (17.4), Mississippi (16.1), Florida (15.4), and Louisiana (15.4) are clear outliers on the high end.

Assault Distribution: Assault rates range from 45 to 337 arrests per 100,000. The distribution is spread out with multiple peaks, showing significant variation between states. The average assault rate is approximately 171 arrests per 100,000.

Urban Population Distribution: Urban population percentages range from 32% (Vermont) to 91% (California). The distribution is fairly even, with most states falling between 50% and 80% urban.

Rape Distribution: Rape rates range from 7.3 to 46.0 arrests per 100,000. Most states are concentrated between 10 and 25, with Nevada (46.0) and Alaska (44.5) standing out as significant outliers.

Visualisation 2: Top 10 States by Crime Type (Bar Charts)

Bar charts were created to identify which states have the highest rates for each crime category.

Top 10 Murder Rates: Georgia (17.4), Mississippi (16.1), Florida (15.4), Louisiana (15.4), South Carolina (14.4), Alabama (13.2), Tennessee (13.2), North Carolina (13.0), Texas (12.7), Nevada (12.2)

Top 10 Assault Rates: North Carolina (337), Florida (335), Maryland (300), Arizona (294), New Mexico (285), South Carolina (279), California (276), Alaska (263), Mississippi (259), Michigan (255)

Top 10 Most Urban States: California (91%), New Jersey (89%), Rhode Island (87%), New York (86%), Massachusetts (85%), Hawaii (83%), Illinois (83%), Nevada (81%), Arizona (80%), Florida (80%)

Top 10 Rape Rates: Nevada (46.0), Alaska (44.5), California (40.6), Colorado (38.7), Michigan (35.1), New Mexico (32.1), Florida (31.9), Arizona (31.0), Oregon (29.3), Missouri (28.2)

Visualisation 3: Correlation Analysis (Heatmap)

A correlation heatmap was created to understand how different crime variables relate to each other.

The correlation matrix shows the following key relationships:

- Murder and Assault (0.80) – Strong positive correlation. States with high murder rates almost always have high assault rates.
- Assault and Rape (0.67) – Moderate-strong positive correlation.
- Murder and Rape (0.56) – Moderate positive correlation.
- UrbanPop and Rape (0.41) – Moderate positive correlation.
- UrbanPop and Assault (0.26) – Weak positive correlation.
- UrbanPop and Murder (0.07) – Very weak positive correlation.

Key insight: The most surprising finding is that urban population percentage has almost no relationship with murder rates (0.07). This suggests that whether a state is highly urban or mostly rural tells us very little about its murder rate.

Visualisation 4: Scatter Plot (Murder vs Assault Relationship)

A scatter plot was created to visually confirm the relationship between murder and assault rates. Each point represents a US state, with point size representing rape rate and point color representing urban population percentage.

Key observations:

- Strong positive relationship confirmed – As murder rates increase, assault rates also increase, forming a clear diagonal pattern from bottom-left to top-right.
- Outlier states identified – Georgia, Mississippi, Louisiana, and South Carolina appear in the top-right corner, confirming they have both high murder and high assault rates.
- Rape rates – States with high murder and assault tend to have larger circles, indicating higher rape rates as well.
- Urban population – No clear pattern emerges, supporting the weak correlation (0.07) between urban population and murder rates.

Summary of Key Findings

After completing this exploratory data analysis on the US Arrests dataset, the following conclusions can be drawn:

Finding 1: The dataset is completely clean with no missing values, no duplicate rows, and correct data types.

Finding 2: Crime rates vary widely across the United States. Georgia has the highest murder rate (17.4), North Carolina has the highest assault rate (337), California is the most urban state (91%), and Nevada has the highest rape rate (46.0).

Finding 3: Violent crimes tend to cluster together. The correlation between murder and assault is 0.80, indicating a very strong positive relationship.

Finding 4: Urban population percentage is not a good predictor of violent crime. The correlation between UrbanPop and murder is only 0.07, meaning rural states can have just as high crime rates as urban states.

Finding 5: Nevada and Alaska are significant outliers for rape rates, both exceeding 40 arrests per 100,000 while most states are below 25.

Finding 6: Southern states (Georgia, Mississippi, Florida, Louisiana, South Carolina, North Carolina) dominate the top ranks for violent crime rates.

This report was written by: **Nomfundo Mpumelelo Sikosana**